Data Structures and Algorithm (PCC-CS301)

Course Details

Credit Points: 3

Pre-Requisites

1. CS 201 (Basic Computation and Principles of C)

2. M101 & M201 (Mathematics), basics of set theory

Syllabus Outline

Unit 1

- Introduction: Basic Terminologies: Elementary Data Organizations
- Data Structure Operations: insertion, deletion, traversal etc.
- Analysis of an Algorithm
- Asymptotic Notations
- Time-Space trade off
- Searching: Linear Search and Binary Search Techniques and their complexity analysis

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Unit 2

- **Stacks and Queues**
  ADT Stack and its operations: Algorithms and their complexity analysis
- Applications of Stacks: Expression Conversion and evaluation – corresponding algorithms and complexity analysis
- ADT queue
- Types of Queues: Simple Queue, Circular Queue, Priority Queue
- Operations on each type of Queues: Algorithms and their analysis

Unit 3

- **Linked Lists**
  Singly linked lists: Representation in memory
- Algorithms of several operations: Traversing, Searching, Insertion into, Deletion from linked list
- Linked representation of Stack and Queue
- Header nodes
- Doubly linked list: operations on it and algorithmic analysis
- Circular Linked Lists: all operations their algorithms and the complexity analysis.

- **Trees**
  Basic Tree Terminologies
- Different types of Trees: Binary Tree
- Threaded Binary Tree
- Binary Search Tree
- AVL Tree
- Tree operations on each of the trees and their algorithms with complexity analysis
- Applications of Binary Trees
- B Tree
- B+Tree: definitions, algorithms and analysis.

Unit 4

- **Sorting and Hashing**
  Objective and properties of different sorting algorithms: Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Merge Sort, Heap Sort
- Performance and Comparison among all the methods
- **Hashing**
- **Graph**
  Basic Terminologies and Representations
- Graph search and traversal algorithms and complexity analysis.

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Data Structures and Algorithm Lab (PCC-CS391)

Laboratory Experiments

- Implementation of array operations
- Stacks and Queues: adding, deleting elements; Circular Queue: Adding & deleting elements
- Merging Problem: Evaluation of expressions; operations on Multiple stacks & queues
- Implementation of linked lists: inserting, deleting, inverting a linked list; Implementation of stacks & queues using linked lists
- Polynomial addition, Polynomial multiplication
- Recursive and Non-recursive traversal of Trees
- Threaded binary tree traversal. AVL tree implementation
- Application of Trees. Application of sorting and searching algorithms
- Hash tables implementation: searching, inserting and deleting, searching & sorting techniques.